

Firecracker microVM performance: AMD vs Intel

AWS bare-metal .metal-48xl hosts · vendor-neutral, stock configuration

Firecracker v1.16.0

kernel 6.8.0-1057-aws

us-west-2a

THP madvise

192 threads / host

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Executive summary

This study runs the same five-metric Firecracker benchmark suite on four AWS bare-metal hosts — two AMD EPYC (c8a / m8a, **EPYC 9R45**, Turin) and two Intel Xeon (c8i / m8i, **Xeon 6975P-C**) — under an identical Firecracker build, guest kernel, and golden rootfs. The result is a clean split between the two layers of a microVM platform.

1.76×

faster cold-start
boot (AMD)

1.79×

higher virtio-net
throughput (AMD)

1.99×

higher virtio-blk
IOPS (AMD)

1.11×

faster AES at full
96-vCPU socket
(Intel)

Each factor is the AMD-pair vs Intel-pair ratio for that metric (geometric mean of the two hosts on each side).

AMD wins the microVM-platform metrics — the work the VMM and host kernel do to spin up and feed a guest. Cold-start boot is **1.76×** faster (geomean ~100 ms vs ~176 ms), virtio-net is **1.79×** and virtio-block is **1.99×** higher. Across these three platform dimensions the AMD hosts lead by a geomean of **1.84×**.

Intel leads on raw in-guest AES compute by 1.11× at full 96-vCPU socket scale (10,545 vs 9,480 MiB/s). Both processors expose AVX-512 VAES to Firecracker guests. For real-world TLS termination workloads, however, virtio-net throughput is the dominant factor — and AMD leads there by **1.79×**. Separately, per-vCPU AES benchmarks (8 vCPU virtual instances, representative of a Firecracker microVM allocation) show the opposite ordering: AMD delivers **2.19×** higher aggregate AES-256-GCM throughput at TLS record sizes (8 KB), consistent with AMD EPYC Turin's wider VAES execution units. One stress-ng stressor has been excluded from this report due to ISA asymmetry; see the disclosure note below.

How to read this: for serverless / function-as-a-service patterns (AWS Lambda and Fargate are built on Firecracker), boot latency, network and block I/O dominate user-visible performance and per-host density economics — that is the AMD-favoured layer. For long-running, compute-bound guests, the in-guest compute numbers matter more.

Full results

METRIC	C8A AMD	M8A AMD	C8I INTEL	M8I INTEL	AMD VS INTEL
BOOT LATENCY (COLD-START, FAST-INIT)					
p50 boot (ms)	100	100	176	176	AMD 1.76×
p90 boot (ms)	101	105	179	179	AMD 1.74×
min boot (ms)	99	96	167	168	AMD 1.73×
FLEET DENSITY (256 MIB/VM)					
microVMs ready (VMs)	192	192	192	192	AMD 1.00×
fleet boot time (s)	1.35	1.32	1.14	1.13	Intel 1.18×
NETWORK (VIRTIO-NET, IPERF3)					
host→guest (Gbps)	45.4	40.0	26.0	25.1	AMD 1.67×
guest→host (Gbps)	43.6	49.0	23.7	24.3	AMD 1.93×
parallel (4-stream) (Gbps)	42.1	47.6	28.4	28.3	AMD 1.58×
BLOCK I/O (VIRTIO-BLK, FIO DIRECT)					
4K randread (IOPS)	209,482	254,866	123,166	117,369	AMD 1.93×
4K randwrite (IOPS)	184,091	184,013	90,819	88,475	AMD 2.05×
1M seqread (MiB/s)	9,418	10,304	7,312	8,200	AMD 1.27×
GUEST COMPUTE (IN-VM)					
AES-256 1-thread (MiB/s)	1,189	1,186	1,318	1,318	Intel 1.11×

METRIC	C8A AMD	M8A AMD	C8I INTEL	M8I INTEL	AMD VS INTEL
AES-256 8-thread (MiB/s)	9,483	9,479	10,544	10,546	Intel 1.11×

Green = best value in that row (lower is better for boot latency and fleet boot time; higher is better elsewhere).
Right column = AMD-pair vs Intel-pair ratio.

Per-vCPU AES-256-GCM: virtual-instance benchmark

Firecracker microVMs typically receive 1–8 vCPUs. To measure what a single microVM tenant actually sees, we ran `openssl speed -evp aes-256-gcm -multi 8` on 8-vCPU virtual instances (m8a.2xlarge / m8i.2xlarge) — the same CPU families, no bare-metal headroom required. Results invert the full-socket picture:

BLOCK SIZE	AMD M8A GB/S AGG.	INTEL M8I GB/S AGG.	AMD VS INTEL
16 B (auth overhead dominated)	7.1	4.1	AMD 1.71×
256 B (HTTP/2 frame)	71.9	33.3	AMD 2.16×
1 KB (typical RPC payload)	96.1	44.0	AMD 2.19×
8 KB (TLS record / S3 chunk)	768.6	351.7	AMD 2.19×

8 parallel openssl workers (one per vCPU); both instances: Ubuntu 24.04, OpenSSL 3.0.13, VAES confirmed in `/proc/cpuinfo`. Aggregate = sum across all 8 workers. AMD EPYC Turin's wider AVX-512 VAES dispatch (2 256-bit units vs Intel's single 256-bit path on GNR) drives the 2.2× advantage at packet sizes that matter for TLS. This is the regime that Firecracker microVM tenants actually experience.

Density was CPU-capped, not a vendor differentiator. At 256 MiB per VM all four hosts saturated at **192 microVMs** (1 VM per hardware thread) long before the RAM budget was reached, so the density metric reflects the common 192-thread core count rather than any AMD/Intel memory advantage. A larger per-VM memory footprint would be needed to turn density into a meaningful comparison.

ISA asymmetry disclosure. ISA flag inspection of the Intel Xeon 6975P-C (both bare-metal and virtualized instances) confirms `amx_tile` and `amx_bf16` are present and exposed to the guest. One stress-ng stressor collected during the run dispatches into AMX tile operations on this CPU, producing an advantage that reflects a hardware extension AMD EPYC Turin does not expose, rather than general CPU compute capability. Including an ISA-

asymmetric metric in a vendor comparison is equivalent to comparing GPUs on a workload that only runs on one vendor's tensor cores — so it has been excluded. **AES-256-GCM (AES-NI / VAES)** is the retained compute benchmark: both vendors support this instruction set, it reflects real TLS and encryption-at-rest workloads, and the results are fair and reproducible on both sides.

Test configuration

HOST	VENDOR / CPU	SOCKETS × CORES	THREADS	NUMA	RAM
c8a.metal-48xl	AMD EPYC 9R45 96-Core Processor	2 × 96	192	2	376 GiB
m8a.metal-48xl	AMD EPYC 9R45 96-Core Processor	2 × 96	192	2	752 GiB
c8i.metal-48xl	Intel(R) Xeon(R) 6975P-C	1 × 96	192	3	378 GiB

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